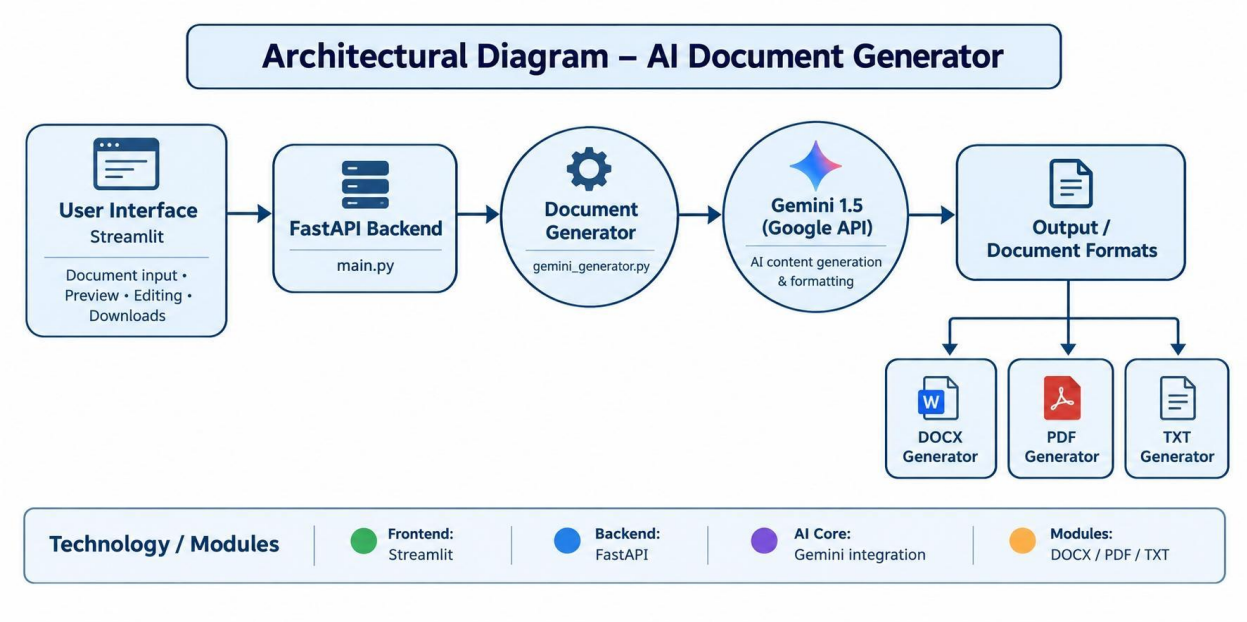


The Core Workflow (Data Flow)

- **User Interface (Streamlit):** The frontend where users interact with the app. It handles document input, file previews, text editing, and downloding the final files.
- **FastAPI Backend (`main.py`):** The backend server that receives requests from the frontend UI and routes them to the appropriate processing modules.
- **Document Generator (`gemini_generator.py`):** The central logic module that prepares the data and communicates with the AI core.
- **Gemini 1.5 (Google API):** The AI core responsible for generating text, creating relevant document content, and handling the structural formatting.
- **Output / Document Formats:** The final stage where the raw AI content is compiled into downloadable files.

2. Export Modules

Once the AI generates the content, the application splits into three distinct exporter modules to generate different file types:

- **DOCX Generator:** Compiles the content into a Microsoft Word document.
- **PDF Generator:** Compiles the content into a standard PDF document.
- **TXT Generator:** Compiles the content into a plain text file.

3. Technology / Modules Color Legend

The bottom panel categorizes the system architecture into a color-coded stack:

- 🟢 **Frontend:** Streamlit
- 🟢 **Backend:** FastAPI
- 🟣 **AI Core:** Gemini integration
- 🟡 **Modules:** DOCX / PDF / TXT formatting scripts

To help tailor this explanation, let me know:

- Are you looking for a **technical code breakdown** on how to connect these specific components (like FastAPI to Gemini)?
- Do you need help **building a specific module** shown in this diagram?